value (linear regression), a category (logistic regression) or a label (clustering).

For all these types of learning, there is one extra step that we didn't have in the optimization problems we have seen so far in this chapter, when we were simply exploring a problem's space: now we also have to choose which model we want to use, which is actually the first thing we need to do.

To better explain this, we'll go into some details of linear regression.

16.4.1 An Example: Linear Regression

Speaking of deceptive names, we couldn't avoid mentioning linear regression; the story of the origin of the name of this learning technique is also fascinating, and it's worth googling it – hopefully we stimulated your curiosity about it.

But what is really important for us is that linear regression ultimately is about finding a model that describes the relation between one or more inputs (aka *independent variables*) and a real number outputted by the model (the *dependent variable*).

We mentioned this "model" a few times now, and you can also see it in figure 16.17: you might legitimately be asking what's that.

The model is a category of mathematical functions that we choose to approximate the true relation between dependent and independent variables in our dataset.

For instance, we might have a dataset associating some characteristics of cars (the year they were built, their engine, how many miles they traveled etc…) to their market price, and we would like to learn the relation between the former and the latter,